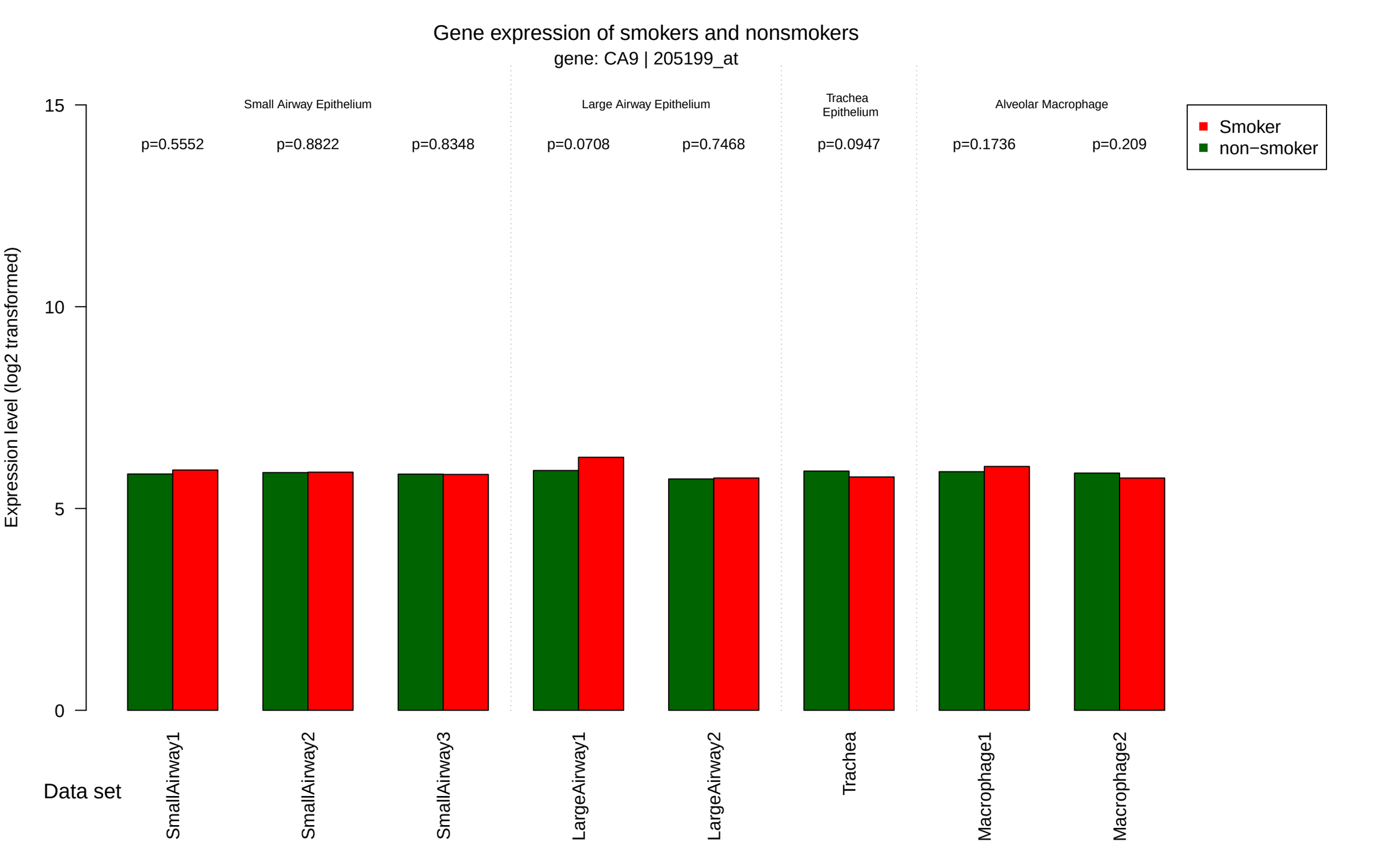
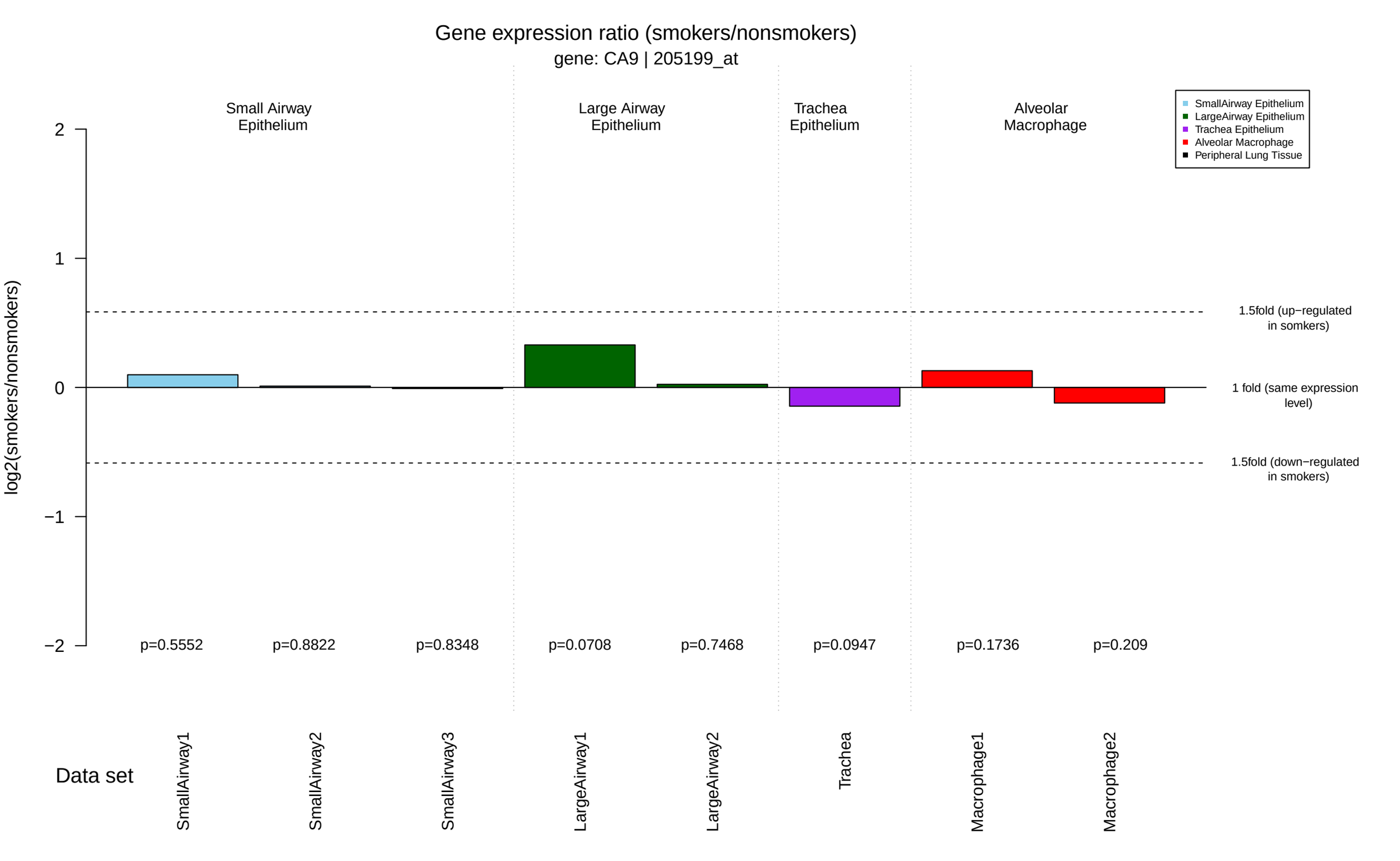
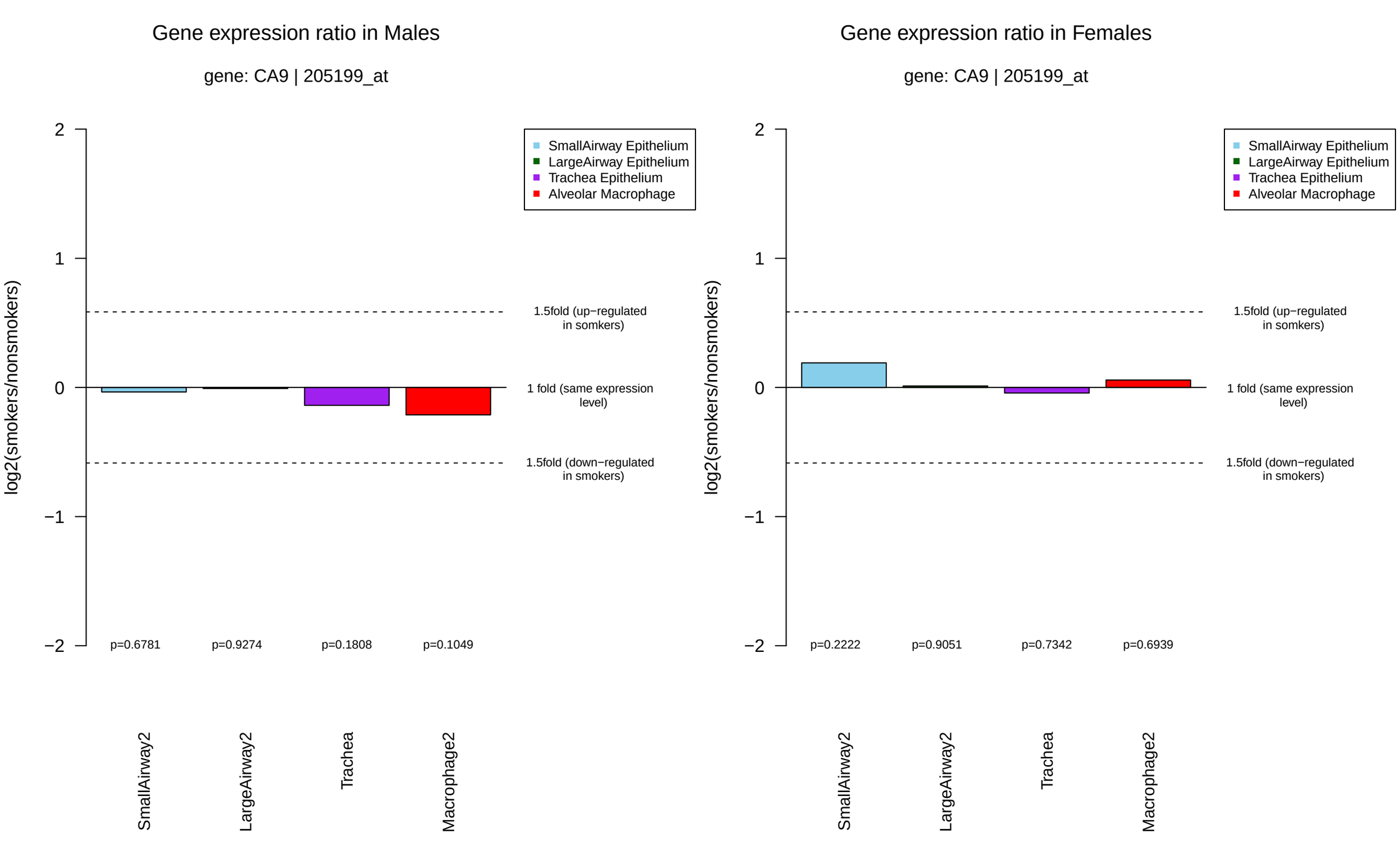
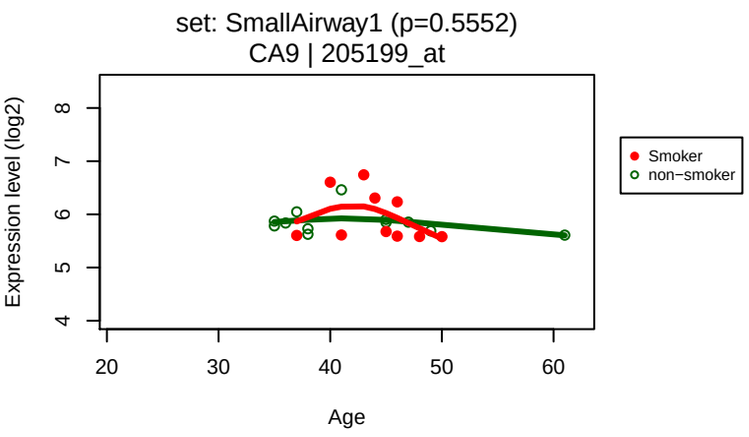
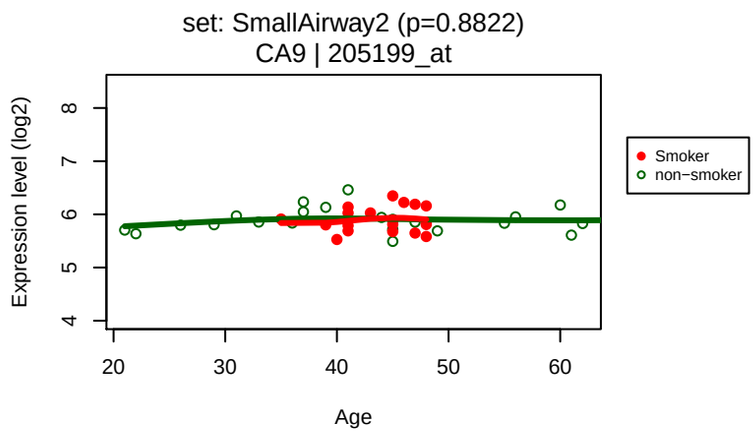
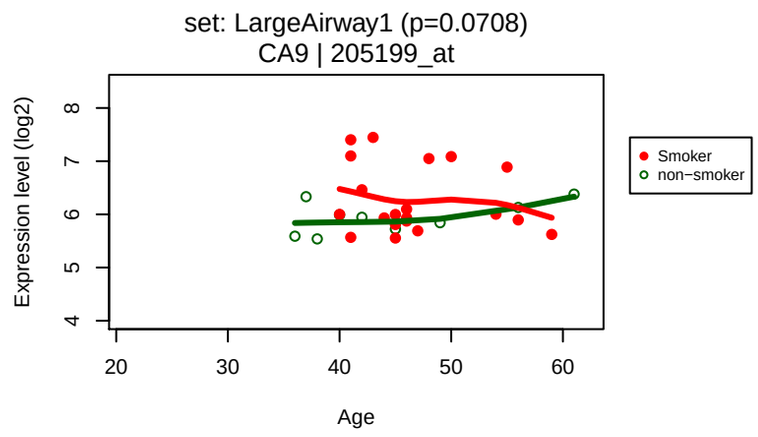

Gene expression of smokers and nonsmokers
gene: CA9 | 205199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5552
p=0.8822
p=0.8348
p=0.0708
p=0.7468
p=0.0947
p=0.1736
p=0.209
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

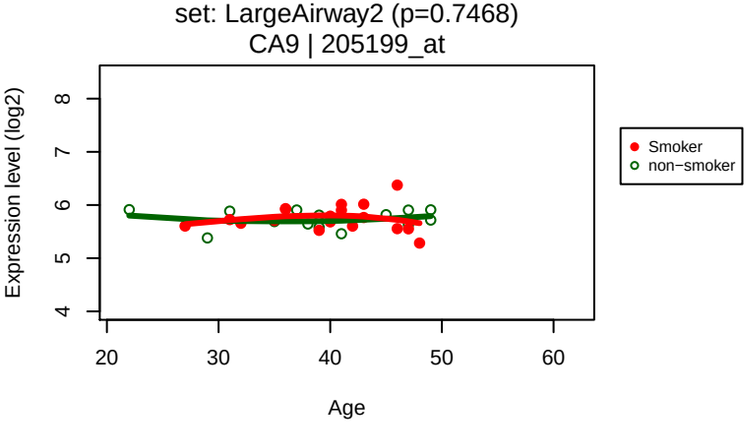

Gene expression ratio (smokers/nonsmokers)
gene: CA9 | 205199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5552
p=0.8822
p=0.8348
p=0.0708
p=0.7468
p=0.0947
p=0.1736
p=0.209
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

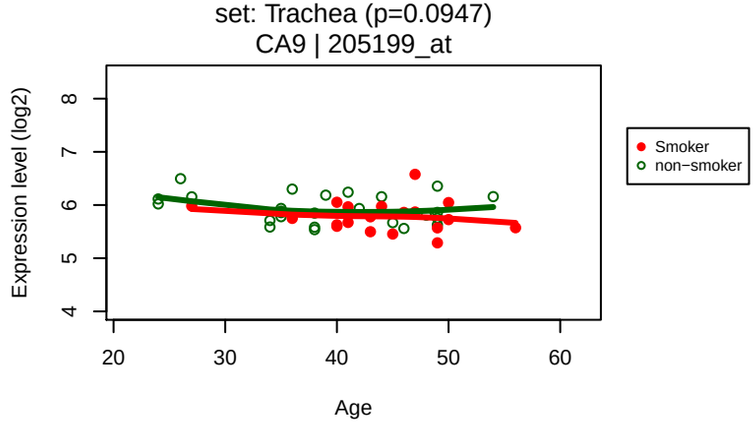

Gene expression ratio in Males
gene: CA9 | 205199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6781
p=0.9274
p=0.1808
p=0.1049
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA9 | 205199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2222
p=0.9051
p=0.7342
p=0.6939
SmallAirway2
LargeAirway2
Trachea
Macrophage2

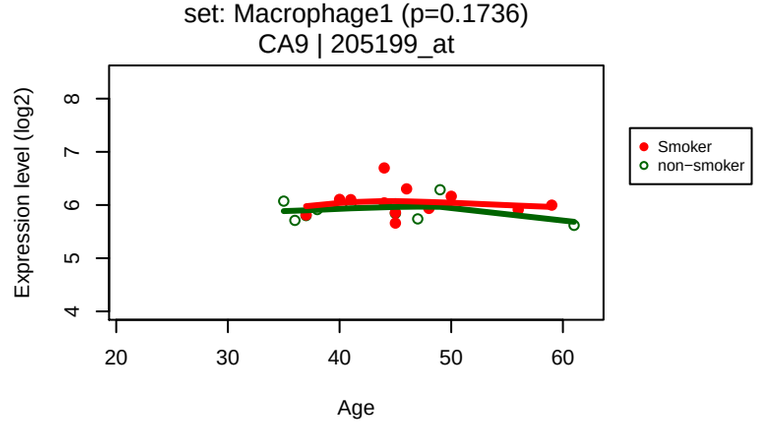

set: SmallAirway1 (p=0.5552)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker

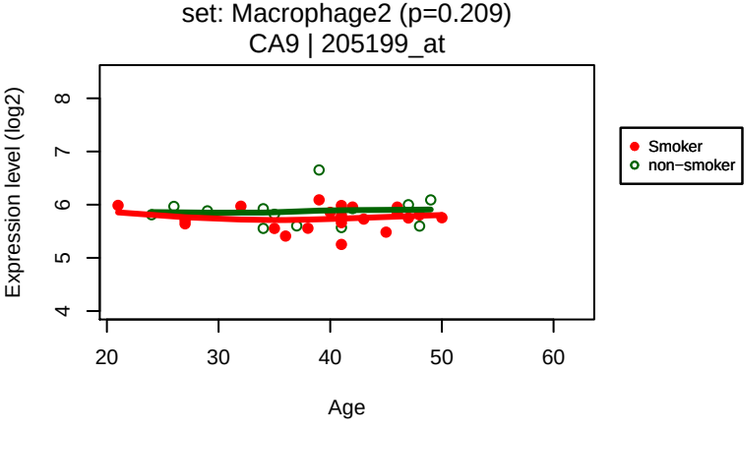

set: SmallAirway2 (p=0.8822)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker

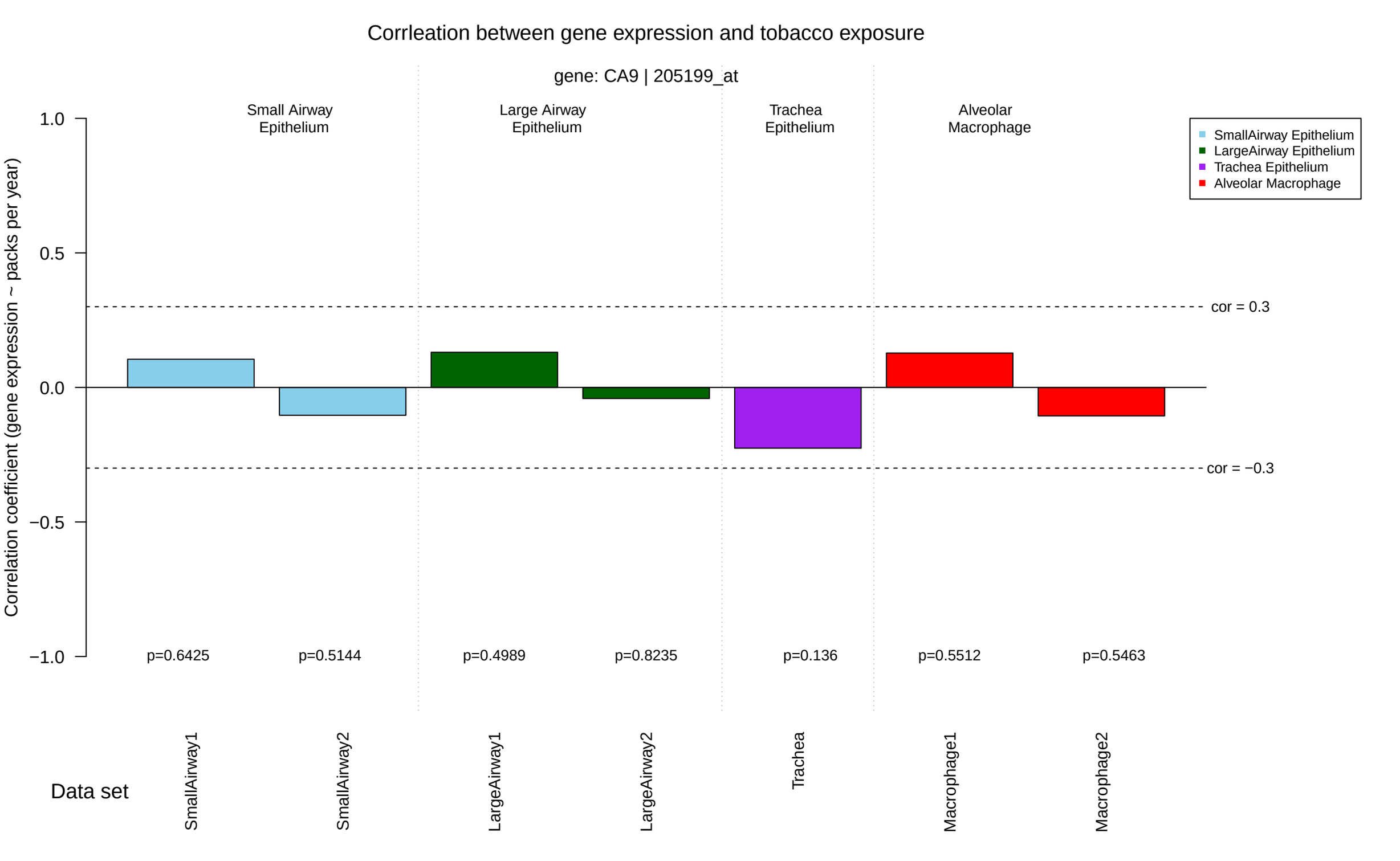

set: LargeAirway1 (p=0.0708)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7468)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0947)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1736)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.209)
CA9 | 205199_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA9 | 205199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6425
p=0.5144
p=0.4989
p=0.8235
p=0.136
p=0.5512
p=0.5463
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2